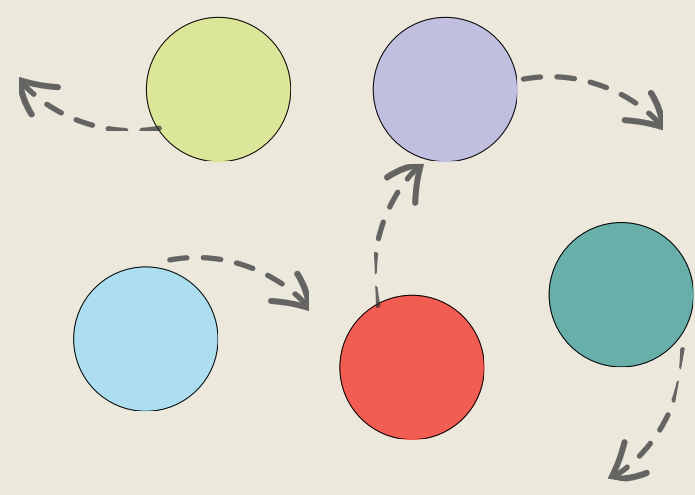


Evaluating the feasibility of a U.S. regulatory sandbox to test mitigations for AI-enabled biological risks

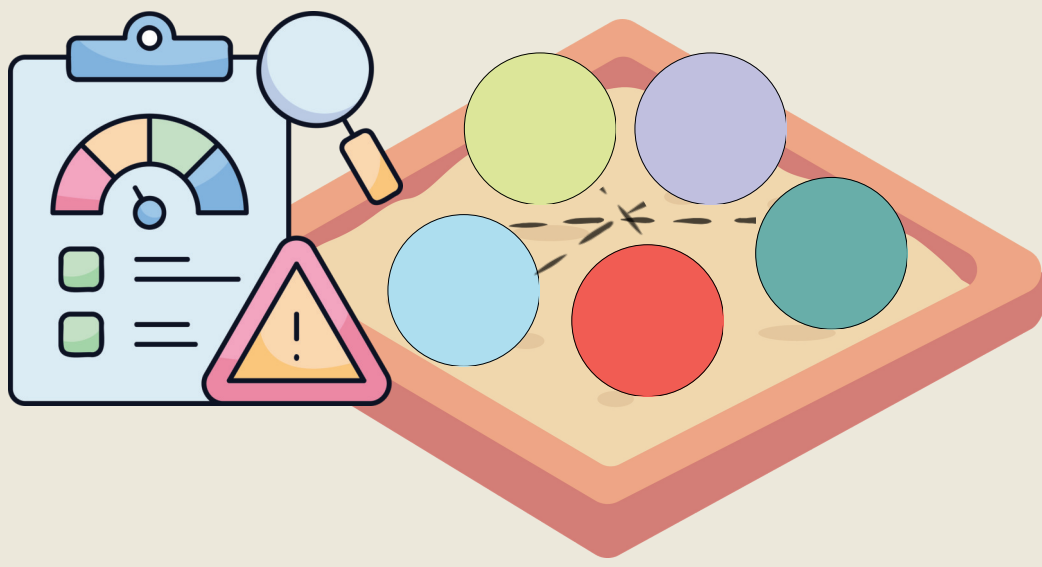
MITIGATIONS EXIST. COORDINATION DOESN'T.

0.1 Theory of Change



THE PROBLEM

Mitigations exist across sectors, but there is no shared way to test if they work



THE IDEA

A sandbox where government, industry, and researchers jointly test mitigations



THE GOAL

Generate evidence on what works, inform policy and adaptation, strengthen mitigations

“ Would a U.S. regulatory sandbox designed to test mitigations for AI-enabled biological risks be feasible and effective, and if so, what institutional design would it require? ”

02. Why This Matters

Artificial intelligence capabilities are advancing faster than the U.S. government’s capacity to assess and mitigate AI-enabled biological risks. Given the national security implications of biological threats, there is a growing need for structured mechanisms to test mitigation strategies in practice.

Expert interviews conducted for this study highlight challenges in the current landscape:

- **Mitigation efforts remain fragmented**, with no central framework guiding the field
- **Mitigation efforts are largely led by private industry**, by default
- **Oversight remains limited**, with no clear statutory authority or coordinated mechanism across agencies

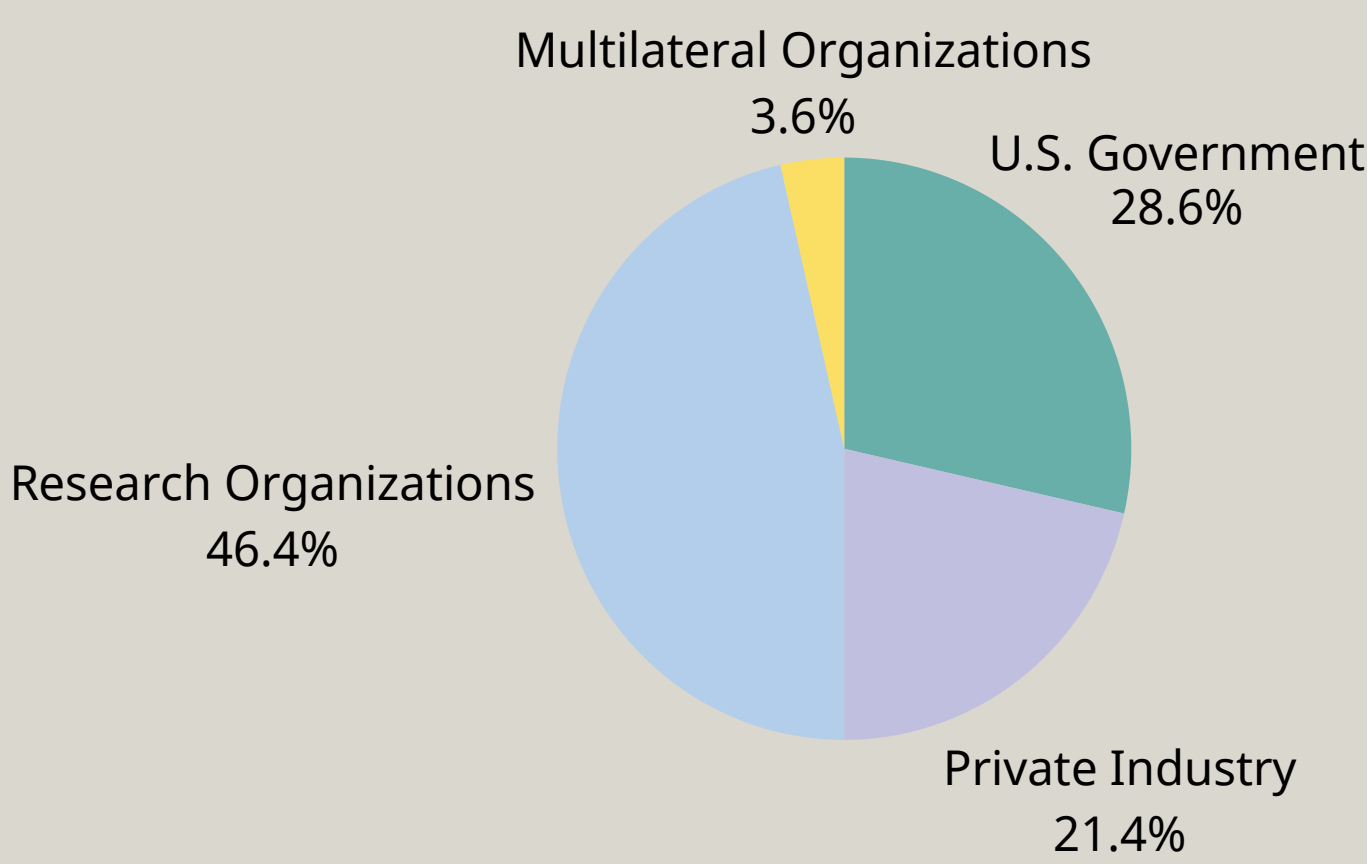
This project examines whether a U.S. regulatory sandbox could address this gap by providing a structured, coordinated way to test whether mitigations are practical, effective, and implementable in real-world settings, particularly for risks related to the intentional misuse of closed-source large language models by non-state actors.

04. What We Did

Study Design: Exploratory policy research using qualitative methods and expert elicitation

Data Collection: 28 semi-structured interviews (avg 45-60 min) conducted virtually / in person

Sampling: Purposive sampling with snowball referrals across four stakeholder groups



Interview Protocol:

1. Current landscape of mitigation efforts
2. Sandbox concept and initial reactions
3. Stakeholder participation
4. Institutional design
5. Political feasibility
6. Framing and messaging considerations

Analysis: Thematic analysis with initial domains informed by the protocol; themes developed inductively based on patterns across participant responses

03. The Sandbox

In other contexts, regulatory sandboxes have been used as controlled settings where companies test AI technologies under flexible requirements and regulator guidance.

In this project, the regulatory sandbox would not focus on testing AI models themselves. Instead, it would be a mechanism where government, industry, and researchers voluntarily collaborate to evaluate:

- whether mitigations actually work
- how they can be monitored
- what capacity is needed to support them at scale

The goal is to inform U.S. policy on the feasibility of this approach and help strengthen oversight and mitigations of AI-enabled biological risks.

05. What We’re Hearing So Far

The following themes emerged from a preliminary analysis of 7 interviews. Findings may evolve as the full dataset is analyzed.

Sandbox Reactions

Participants broadly supported the sandbox concept, noting it could add value by enabling standardization, coordination, and shared benchmarks across sectors.

Conditions for Failure

Some cautioned the sandbox could fail if it over-focuses on model-level interventions, relies on wrong assumptions about threats, or lacks mechanisms to compel action.

Considerations for Participation

Intellectual property, trade secrets, and reputational risk were cited as top barriers to private industry participation, with IP protection, confidentiality safeguards, and liability protections seen as necessary to enable buy-in.

Institutional Host

Views were mixed on who should host the sandbox. Some favored federal government, particularly NIST; others suggested an industry consortium; and some were uncertain.

Implementation Barriers

Participants did not report significant technical barriers for implementation. Political barriers were more prominent, including the term "sandbox" becoming politically charged and the need for Congressional or executive order authority.

How do YOU envision the sandbox?

Join In! Grab a sticky and share your idea

Who should lead it?

U.S. Government

Private Industry

Research Organization

Other

What should be tested?

What’s the biggest barrier?